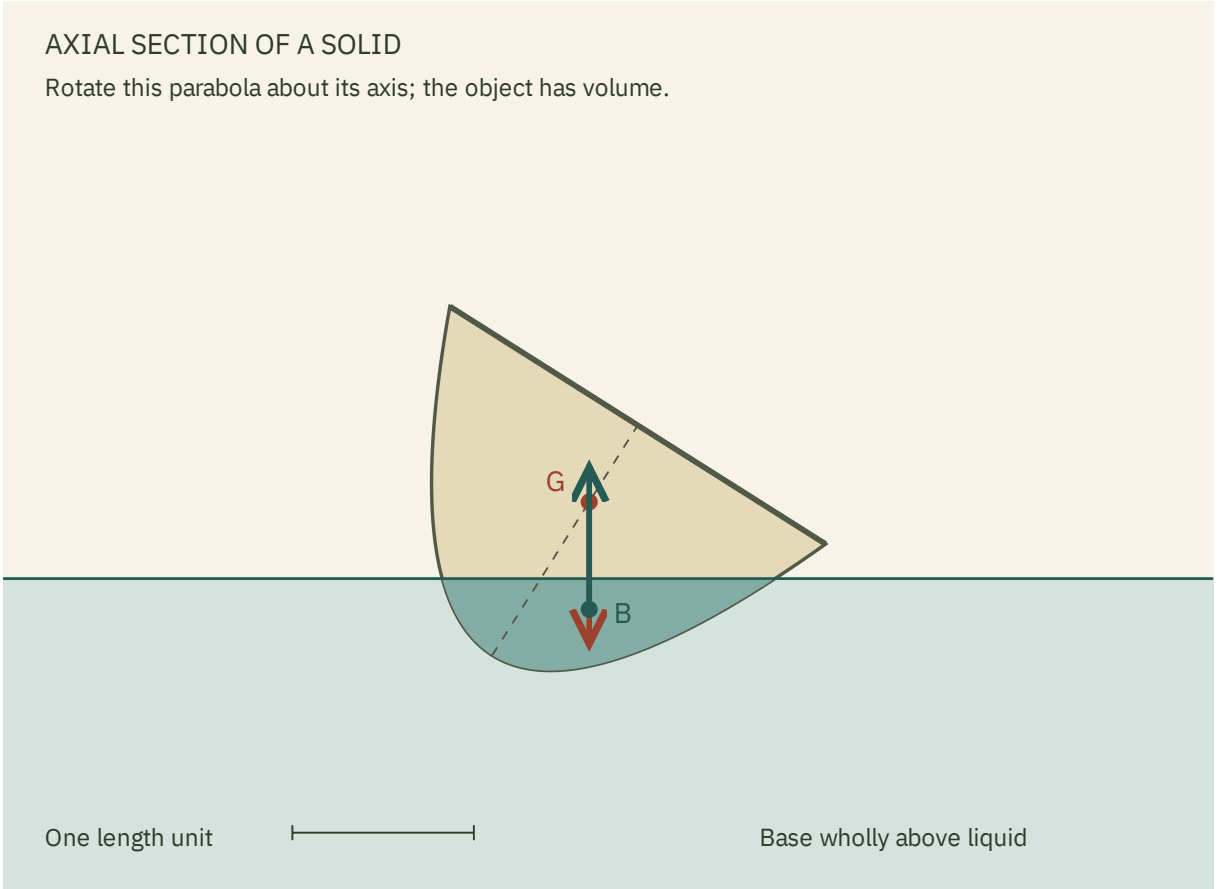
It floats.
Will it stay upright?

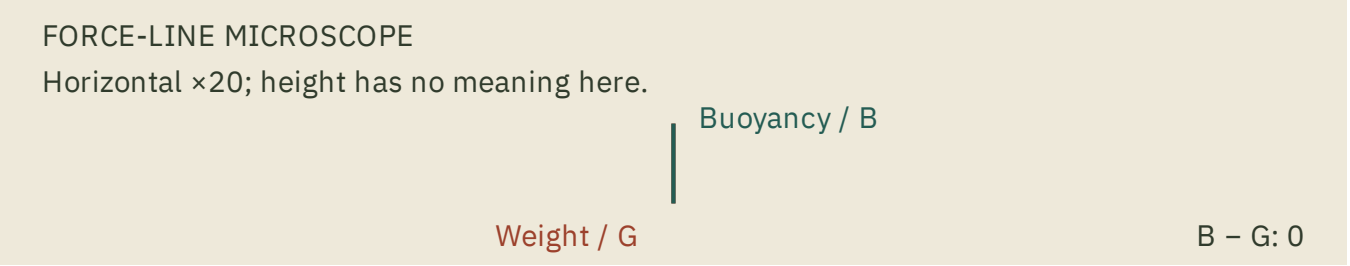
A solid can have equal weight and buoyancy while a small tilt carries it farther from upright. This is a static force experiment, not an animation of physical motion.

Retained settings: axis a = 1.5, solid density = 16% of liquid, tilt = 32.3115332°. Selected reading 6.

Return to this experiment online. The complete reading below uses embedded assets.

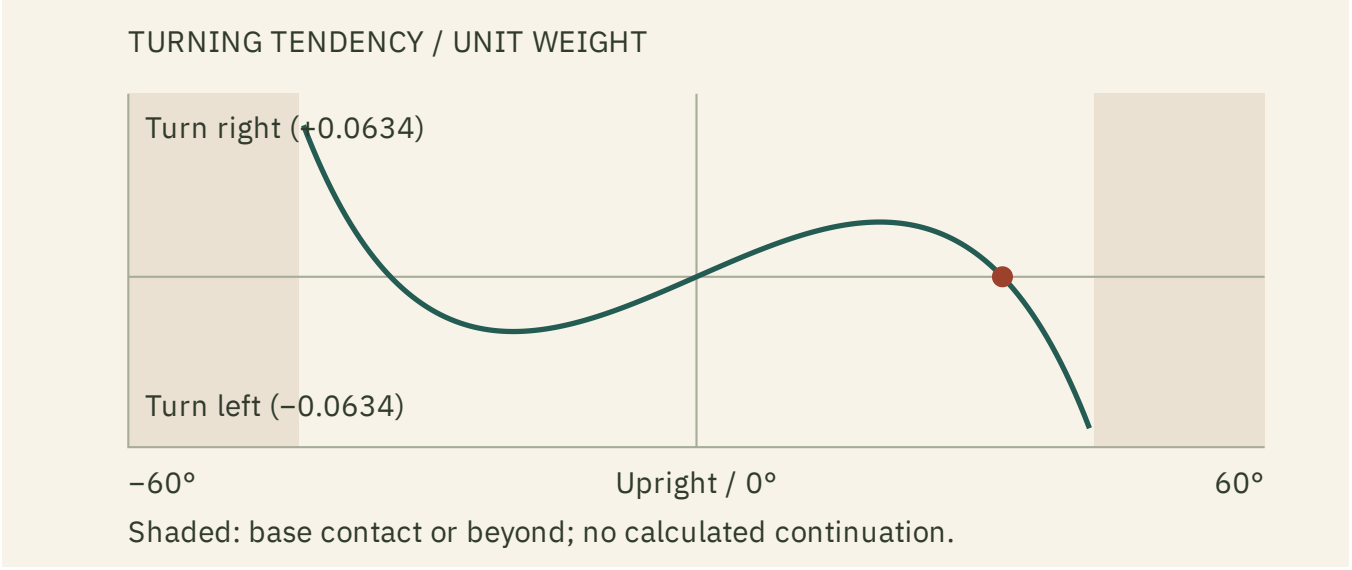


Equal horizontal and vertical length scales. G is the whole solid's centre of gravity; B is the displaced liquid's centre of volume. Arrows have equal lengths to show equal force magnitudes, and their lengths are schematic.



Current experiment · rounded analytic results

Shape / axis a	1.5
Solid / liquid density	16%
Tilt to the right	32.3115332°
Total volume	3.5342917 length units³
Base status	Wholly dry
Displaced volume	0.5654867 length units³
B − G horizontal gap	0 length units
Upright response	unstable
First base-contact tilt	41.9965938° in either direction
Dry-base oblique equilibrium	±32.3115332°



What the modern model assumes

Uniform solid paraboloid; flat, still, unbounded liquid; parallel gravity; base wholly dry. Static forces, not motion or a vessel reconstruction. The solid is the region $x^2 + y^2 \le z \le a$. The broad flat circle is its base. Density σ determines the upright submerged depth $d = a\sqrt{\sigma}$. For tilt θ, $b = \tan\theta$ and the candidate water plane is $z = bx + c$ with $c = d - b^2/4$. The displaced cap's centre is $(b/2, 0, 2d/3 + b^2/4)$; the solid's centre is $(0, 0, 2a/3)$. Their horizontal separation in the world is $\sin\theta\,[1/2 - 2(a-d)/3 + \tan^2\theta/4]$. A positive gap turns the body left; a negative gap turns it right. A zero gap alone says balanced, not stable.

The full displaced cap must fit below $z = a$: $(|b|/2 + \sqrt{d})^2 < a$. At contact or beyond it, the model withholds displaced volume and force results. It neither continues a convenient infinite-paraboloid formula nor solves the base-crossing case. Rounded values and plotted samples are illustrations of the analytic model,

not Archimedes' geometrical proof.

The upright linear stiffness changes sign at $\sigma = (1 - 3/(4a))^2$ when $a > 3/4$. At equality finite nonzero dry-base tilts still restore, although the linear term is zero. Below that density the upright state is balanced but unstable; a nonzero balanced tilt can exist. Its dry-base condition is checked separately.

Readings, witnesses and editorial signs

Greek follows the released Netz–Wilson transcription. Red angle brackets mark supplied letters, dotted underlines uncertain readings, italics expanded abbreviations, and wavy underlines a reading marked sic. These are selected passages with new English teaching translations, not the full treatise or a new decipherment. Latin follows the pre-palimpsest text printed by Heiberg in 1881. The proof break and the missing denominator remain visible. No Latin is invented for the absent proof ending.

A condition on the shape

Τὸ ὀρθὸν τμᾶμα τοῦ ὀρθογωνίου κωνοειδοῦς, ὅταν τὸν ἄξονα σχῆι μὴ μείζονα ἢ ἡμιόλιον τῆς μέχρι τοῦ ἄξονος,

The right segment of the right-angled conoid, when it has its axis no greater than one and a half times the line up to the axis, …

Latin in Heiberg's 1881 edition, printed p. 376.

Recta portio rectanguli conoidalis quando axem habuerit non maiorem, quam emiolium eius, quae usque axem,

A right-angled conoid is a paraboloid of revolution, not a cone with straight sides. In our coordinates $z = x^2 + y^2$, the source's 'line up to the axis' is 1/2. Thus the axis condition becomes $a \le 3/4$. 'Right segment' describes its perpendicular cut; the solid can then be tilted in the liquid.

τμᾶμα — a segment. Doric form of τμῆμα; a part cut from a solid. Here it includes its volume.

ὀρθογωνίου κωνοειδοῦς — of a right-angled conoid. The genitive names the parent solid, a paraboloid of revolution. It is not a straight-sided cone.

ἄξονα — axis. Accusative after σχῆι, 'has'. It is the vertex-to-base length a.

ἡμιόλιον — one and a half times. A whole plus a half: 3:2, not one half. Latin emiolium transliterates the Greek.

τῆς μέχρι τοῦ ἄξονος — the [line] up to the axis. A technical parabola parameter; 1/2 in $z = x^2$. Its value changes with coordinate convention.

Any ratio, with a dry base

πάντα λόγον ἔχον πρὸς τὸ ὑγρὸν τῶι βάρει, ἀφεθὲν εἰς τὸ ὑγρὸν οὕτως, ὥ⟨σ⟩τε τὴν βάσιν αὐτοῦ μὴ ἅπτεσθαι τοῦ ὑγροῦ,

… having any ratio in weight to the liquid, released into the liquid in such a way that its base does not touch the liquid, …

Latin in Heiberg's 1881 edition, printed p. 376.

omnem proportionem habens ad humidum in grauitate, dimissa in humido ita, ut basis ipsius non tangat humidum,

The base is the flat circular cut at the top, not the pointed vertex. The surrounding argument concerns a body lighter than an equal volume of liquid. 'Any ratio' does not license a solid denser than the liquid to float. The workshop explores densities from 4% to 90% of the liquid's.

πάντα λόγον — every ratio. Accusative object of ἔχον. The lighter-than-liquid context still applies.

τῶι βάρει — in weight. Dative specifying the respect in which the ratio is taken; equal-volume weights give relative density.

βάσιν — base. The circular plane cut at the broad end. It does not mean the pointed bottom of the drawing.

μὴ ἅπτεσθαι — not to touch. μή negates the infinitive. ἅπτεσθαι takes the genitive τοῦ ὑγροῦ, 'the liquid'.

Will not remain tilted

τεθὲν κεκλιμένον οὐ μενεῖ κεκλιμένον, ἀλλὰ ἀποκαταστήσεται ὀρθόν.

… placed tilted, it will not remain tilted, but will be restored upright.

Latin in Heiberg's 1881 edition, printed p. 377 (begins on p. 376).

posita inclinata, non manet inclinata, sed restituetur recta.

The Latin sentence crosses the printed page break: posita inclinata, non manet in- ends p. 376; clinata begins p. 377. We join the line-break word without changing it. The model computes the direction of a static turning moment. It does not animate a measured return, time its motion or model damping.

κεκλιμένον — having been tilted. Perfect passive participle, neuter, agreeing with τμᾶμα. The state after tilting is the question.

οὐ μενεῖ — will not remain. Future of μένω, negated by οὐ. Latin non manet uses a present verb.

ἀποκαταστήσεται — will be restored. Future middle: the conclusion is a return to the upright position.

What upright means

ὀρθὸν δὲ λέγω καθεστηκέναι τὸ τοιοῦτο τμᾶμα, ὁπόταν τὸ ἀποτετμηκὸς αὐτὸ ἐπίπεδον ᾖι *παρὰ* τὴν ἐπιφάνειαν ᾖι τοῦ ὑγροῦ.

I say that such a segment stands upright whenever the plane that cut it off is parallel to the surface of the liquid.

Latin in Heiberg's 1881 edition, printed p. 377.

rectam dico consistere talem portionem, quando quod secuit ipsam fuerit aequidistanter superficiei humidi.

The transmitted Greek repeats ᾖι around 'parallel to the surface'; it remains visible. The Latin's quod secuit ipsam, 'what cut it', is its compressed counterpart to the Greek plane. In the flat-surface model, a horizontal base and a vertical axis describe the same upright orientation.

ἐπίπεδον — plane. The cutting plane is parallel to the liquid surface when the body is upright.

Reading 5 / II.2 / end of proof / Arch08r

The proof continues in Greek

δῆλον, ⟨ὡ⟩ς οὐ με⟨ν⟩εῖ τὸ τμᾶμ⟨α⟩ ⟨ἐ⟩ν τῶι ὑγρῶι ἀλλὰ τὰ μὲν κατὰ τὸ Α εἰς τὸ ἄνω ἐνεχθήσεται, τὰ δὲ κατὰ τὸ Λ εἰς τὸ κάτω, ἀεὶ ἔστεν, ἕως ἂν ὀρθὸν ἀποκατασταθῆι.

It is clear that the segment will not remain in the liquid [in this position]: the parts at A will be carried upward, and those at Λ downward, continuously, until it is restored upright.

No corresponding Latin in the retained 1881 witness; its proof breaks off earlier.

The 1881 Latin witness ends its II.2 proof after drawing a tangent. Its footnote says the last part is missing in Tartaglia and Commandino supplied his own. This Greek conclusion comes from the later released palimpsest transcription. There is no corresponding Latin excerpt in the retained 1881 page. The source's A and Λ label its figure, not the workshop's G and B. The transcription's odd ἀεὶ ἔστεν remains marked as sic; the English renders the surrounding conclusion.

τὰ μὲν … τὰ δὲ — the parts on one side … the parts on the other. A paired contrast: one side goes up, the other down. It describes rotation.

A taller solid needs a density condition

ΤΟ ὀρθὸν τμᾶμα τοῦ ὀρθογωνί*ου* κωνοειδοῦς, ὁπόταν κουφότερον ᾖ τοῦ ὑγροῦ καὶ τὸν ἄξονα ἔχῃ μεῖζον ἡμιόλιον τῆς μέχρι τοῦ ἄξονος, ἐὰν τῶι βάρει πρὸς τὸ ἴσογκον ὑγρὸν μὴ ἐλάσ⟨σονα⟩ ⟨λόγον⟩ ⟨ἔχη⟩ ⟨τοῦ⟩ ⟨ὃν⟩ ⟨ἔχει⟩ ⟨τὸ⟩ ⟨τετράγωνον⟩ ⟨τὸ⟩ ⟨ἀπὸ⟩ ⟨τῆς⟩ ⟨ὑπεροχ⟩ῆς, ἣ μεῖζόν ἐστιν ⟨ὁ⟩ ⟨ἄξ⟩ων ἢ ἡμιόλιος ⟨τῆς⟩ ⟨μέ⟩χ⟨ρι⟩ τοῦ ⟨ἄξ⟩ον⟨*ος*⟩, ⟨πρὸς⟩ ⟨τὸ⟩ ⟨τετράγωνον⟩ ⟨τὸ⟩ ⟨ἀπὸ⟩ ⟨τοῦ⟩ ⟨ἄξ⟩ονος, ⟨ἀφεθὲ⟩ν ε⟨ἰ⟩ς τ⟨ὸ⟩ ⟨ὑγ⟩ρὸν ⟨οὕτως⟩, ⟨ὥσ⟩τ⟨ε⟩ τὴν ⟨β⟩άσι⟨ν⟩ α⟨ὐτοῦ⟩ ⟨μὴ⟩ ⟨ἅπτεσθαι⟩ ⟨τοῦ⟩ ⟨ὑγροῦ⟩, ⟨τεθὲν⟩ κ⟨εκ⟩λ⟨ιμέ⟩νο⟨ν⟩ ο⟨ὐ⟩ ⟨μενεῖ⟩ ⟨κεκλιμέν⟩ον, ⟨ἀ⟩λλ⟨ὰ⟩ ἀπο⟨κ⟩α⟨τα⟩στ⟨ήσεται⟩ ⟨εἰς⟩ ὀρθό⟨ν⟩.

The right segment of the right-angled conoid, whenever it is lighter than the liquid and has its axis greater than one and a half times the line up to the axis, if its ratio in weight to an equal volume of liquid is no less than the ratio of the square on the excess, by which the axis exceeds one and a half times the line up to the axis, to the square on the axis, released into the liquid so that its base does not touch the liquid, placed tilted, will not remain tilted but will be restored upright.

Latin in Heiberg's 1881 edition, printed p. 379–380; denominator omitted in the body text.

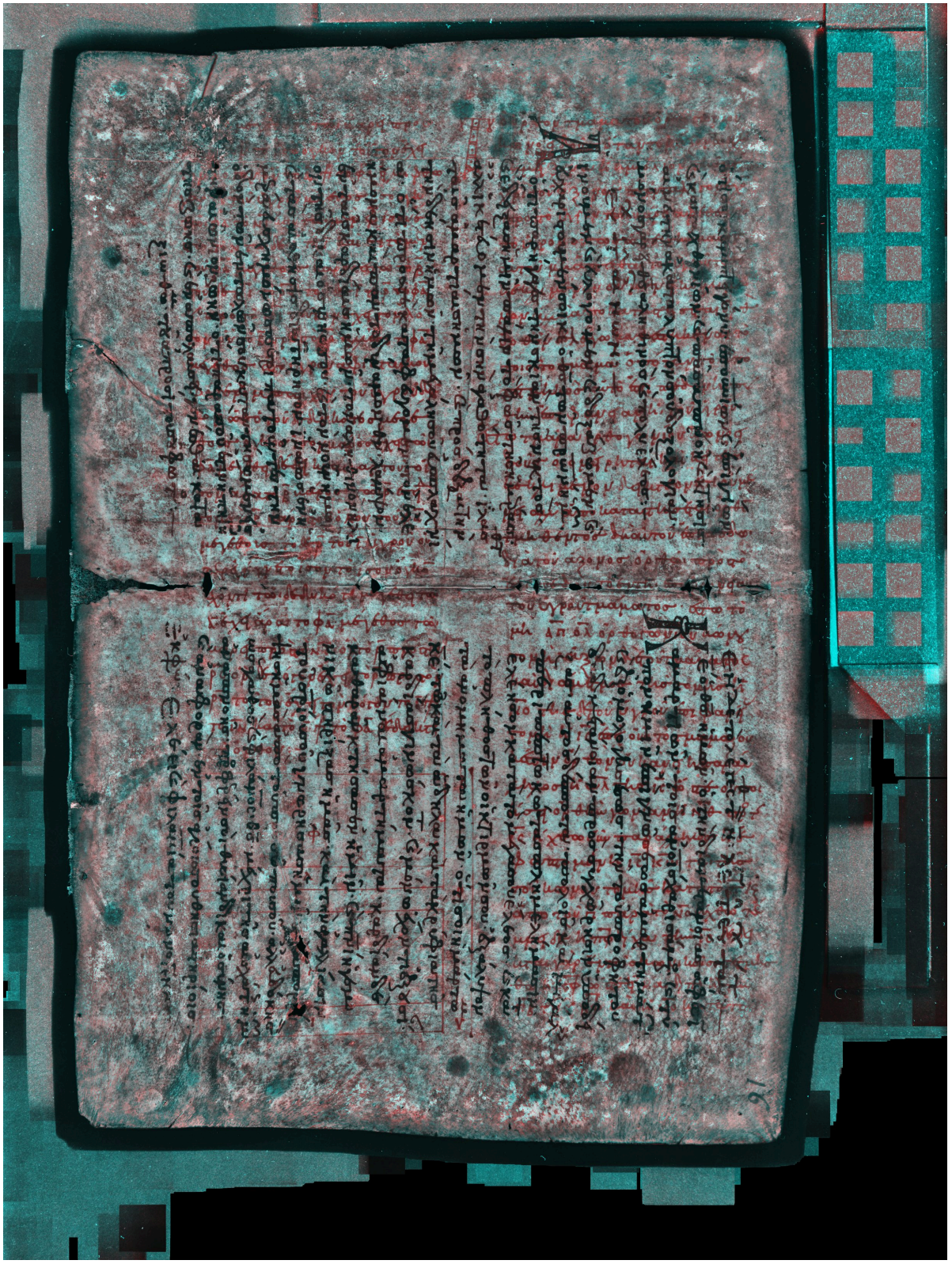
Recta portio rectanguli conoidalis quando fuerit leuior humido et axem habuerit maiorem quam emiolium eius, quae usque ad axem, si in grauitate ad humidum aequae molis non minorem proportionem habeat illa, quam habet tetragonum quod ab excessu, quo maior est axis quam emiolius eius, quae usque ad axem, dimissa in humido ita, ut basis ipsius non tangat humidum, posita inclinata, non manet inclinata, sed restituetur in rectum.

In modern notation the Greek density condition is $\sigma \geq ((a - 3/4)/a)^2$. The Latin printed across pp. 379–380 omits the denominator after 'axis'; Heiberg's p. 380 footnote says to add ad tetragonum quod ab axe. We retain the body text and show that footnote, rather than silently repairing the Latin. Equality cancels the linear stiffness; finite dry-base tilts still restore. Many Greek letters in this statement are supplied by the editors, as the red brackets show.

ἴσογκον ὑγρόν — an equal volume of liquid. The comparison fixes volume before comparing weights. A small dense object can weigh less than a large light one.

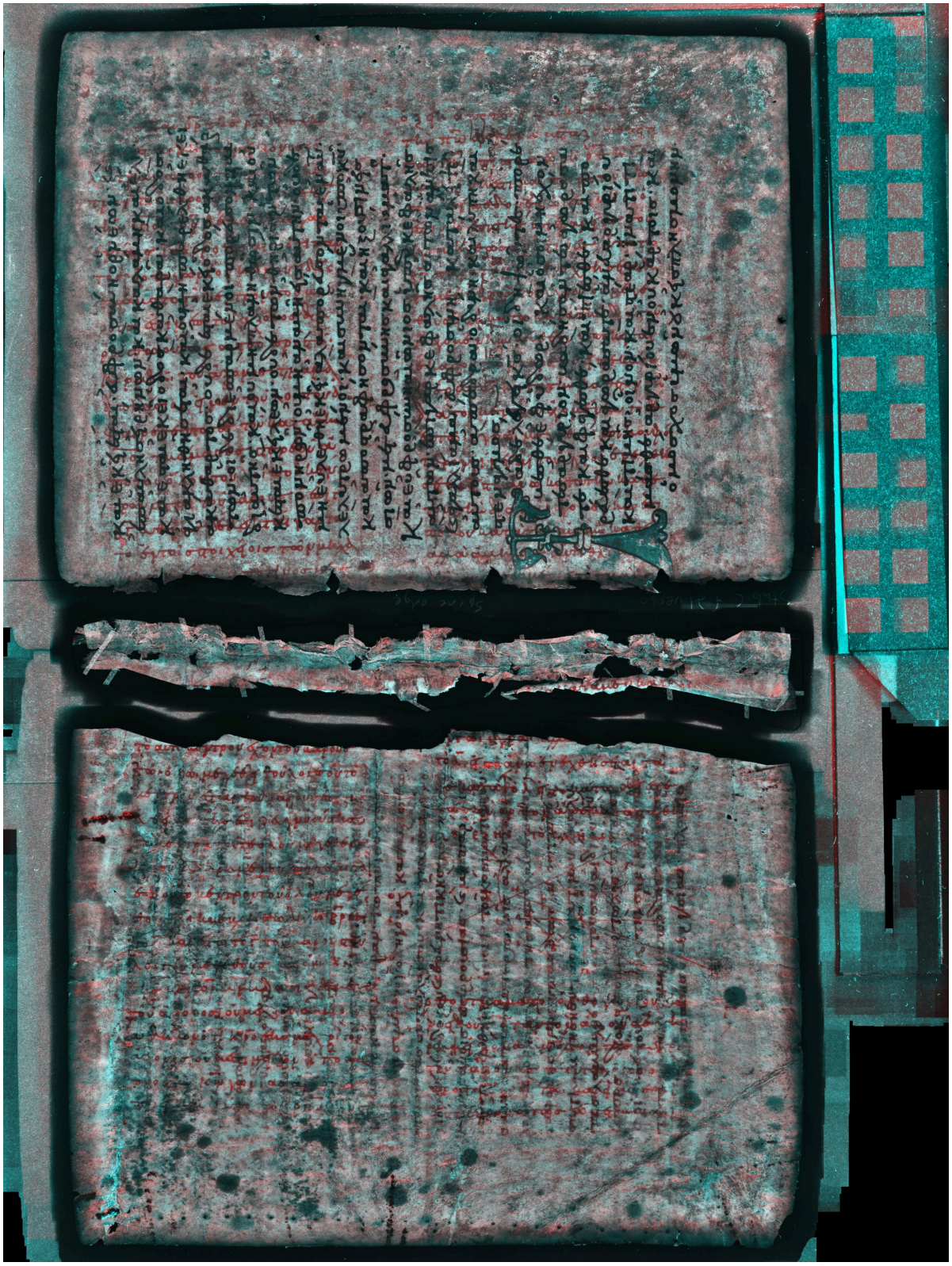
μὴ ἐλάσσονα λόγον — a ratio no smaller. An inclusive lower bound. Equality belongs to the hypothesis; the workshop retains that boundary case.

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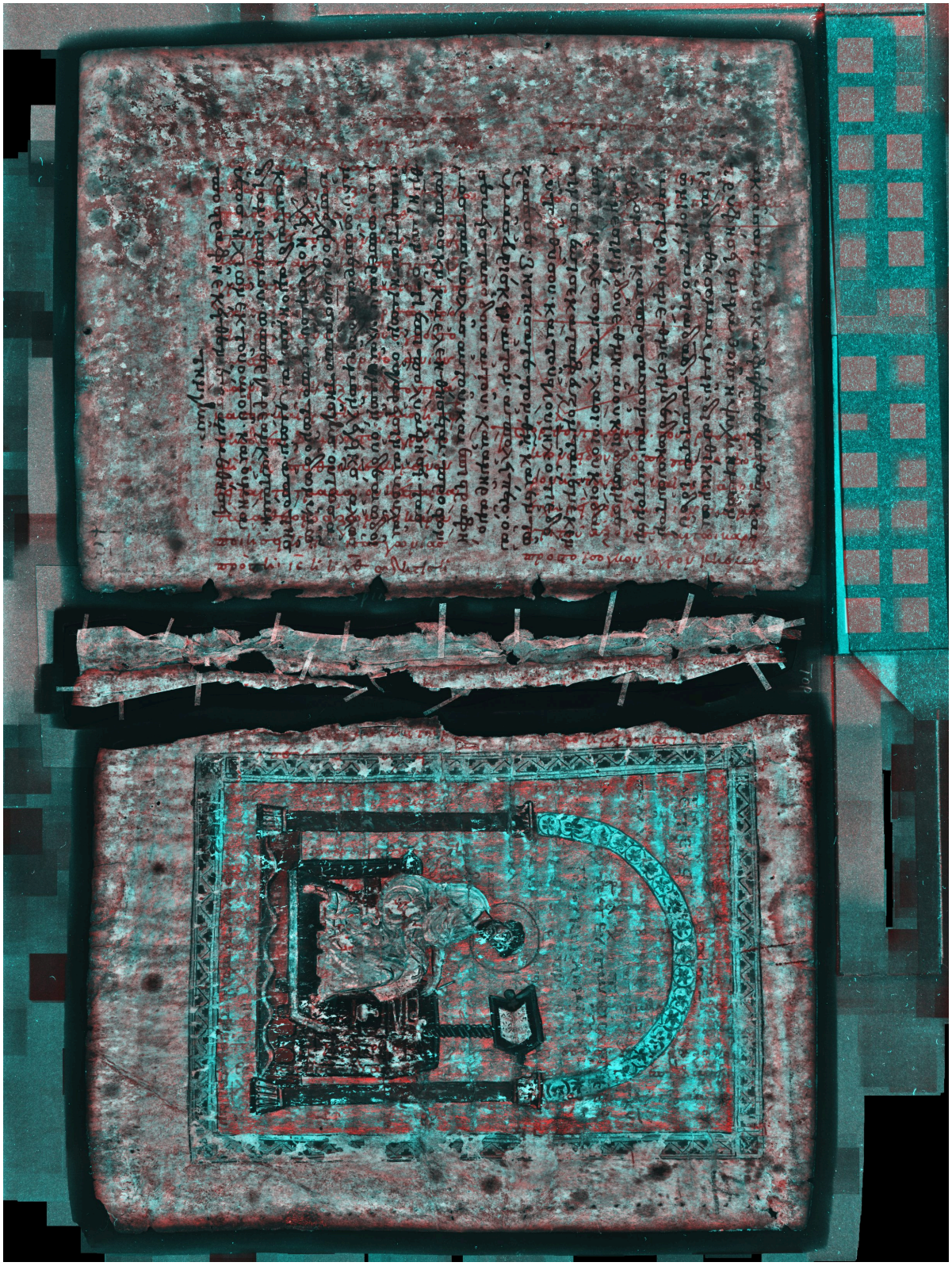
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1881 Latin · printed p. 376 · public domain

existens humido, palam, quod demersae magnitudinis moles humidi habet grauitatem aequalem cum magnitudine *FA*. demonstratum est enim hoc [I, 5]. et quoniam quod secundum *A* humidum est *I*, ipsius autem *I* grauitas est *R*, ipsius autem *FA* grauitas est *B*, grauitas *B*, quae est habentis aequalem molem totius magnitudinis *FA*, est aequalis grauitati humidi *I*, scilicet ipsi *R*; et quoniam est, ut magnitudo *FA* ad humidum, quod secundum ipsam, scilicet *NI*, ita *B* ad *RO*, aequale autem est *B* ipsi *R*, ut autem *R* ad *RO*, ita *I* ad *NI* et *A* ad *FA*, ut ergo *FA* ad humidum, quod secundum ipsam, in grauitate, magnitudo *A* ad *FA*. †factum est aequale demersae magnitudinis, scilicet *A*. habet ergo magnitudo *FA* in grauitate ad *NI*, ita *B* ad *RO*. quam autem proportionem habet *R* ad *RO*, hanc habet proportionem ... ad *R*, ... et *A* ad *FA*. demonstratum est enim.

II.

Recta portio rectanguli conoidalis quando axem habuerit non maiorem, quam emiolium eius, quae usque axem, omnem proportionem habens ad humidum in grauitate, dimissa in humido ita, ut basis ipsius non tangat humidum, posita inclinata, non manet in-

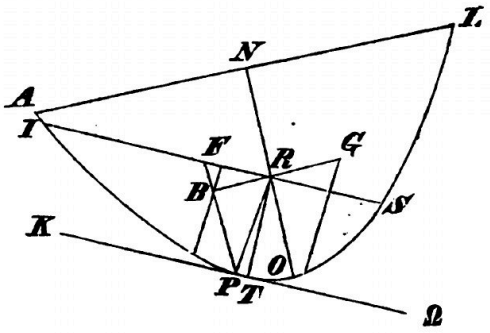
1. „demeraes“ Tartalea; „tantam humidi molem, quanta est pars magnitudinis demersa“ Comm. 4. quod secundum] τὸ κατὰ τὴν *A* ὑγρόν. 6. „aequalitate mole“ Tartalea; „grauitatem aequalem“ Nizzius male. 10. *B*] (prius) „*BO*“ Tartalea. 12. „ipsa“ Tartalea. 13. factum] sequentia uerba sensu carent, nec opus sunt; „quod demonstrare oportebat“ Comm. ceteris omissis. 17. ad *R*] in media lacuna Tartalea. 20. *non maiorem*] Torellius p. XVIII; „maiorem“ Tartalea; „minorem“ Comm.

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clinata, sed restituetur recta. rectam dico consistere talem portionem, quando quod secuit ipsam fuerit aequidistanter superficiei humidi.

sit portio rectanguli conoidalis, qualis dicta est, et iaceat inclinata. demonstrandum, quod non manet, sed restituetur recta. secta autem ipsa plano per



axem recte ad planum, quod in superficie humidi, portionis sectio sit quae $APOL$ rectanguli coni sectio [$\pi\varepsilon\varrho\grave{\iota}$ $\varkappa\omega\nu$. 11], axis autem portionis et diameter sectionis quae NO, superficiei autem humidi quae IS. si igitur portio non est recta, non utique erit quae AL aequidistans ipsi IS. quare non faciet angulum rectum quae NO ad IS. ducatur ergo quae $K\Omega$ contingens sectionem coni penes P.†[1])

1) Pars extrema demonstrationis apud Tartaleam deest; de suo adiecit Commandinus.

8. quae $APOL$] „que apol.“ Tartalea. 10. superficiei] pendet ab „sectio“ lin. 8. quae IS] „quam K“ Tartalea. 12. IS] „is K“ Tartalea.

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IS contingens sectionem *APOL* penes *O*, et solidae quidem magnitudinis *APOL* centrum grauitatis sit *R*, ipsius autem *IPOS* solidi centrum *B*, et copulata quae *BR* educatur, et centrum grauitatis reliquae figurae, scilicet *ISLA*, sit *G* [cfr. *ἐπιπ. ἰσοῤῥ.* I, 8]. similiter[1]) demonstrabitur angulus quidem qui sub *ROK* acutus, perpendicularis quae ab *R*, *TR*, ad *KΩ* producitur, cadens inter *K* et *Ω*, sitque *RT*. si autem ab ipsis *G*, *B* ducantur aequedistanter ipsi *RT*, quod quidem in humido absumptum feretur sursum secundum productam per *G* [I hypoth. 2]. quod autem extra humidum secundum productam per *B* feretur deorsum, et non manet solidum *APOL* sic se habens in humido, sed quod quidem secundum *A* habebit lationem sursum, quod autem secundum *L* deorsum, donec fiat quae *PF* secundum perpendicularem.

IV.

Recta portio rectanguli conoidalis quando fuerit leuior humido et axem habuerit maiorem quam emiolium eius, quae usque ad axem, si in grauitate ad humidum aequae molis non minorem proportionem habeat illa, quam habet tetragonum quod ab excessu, quo maior est axis quam emiolius eius, quae usque

1) Sc. ac supra in demonstratione prop. 2, quae intercidit; de re u. Nizze p. 234, *η*.

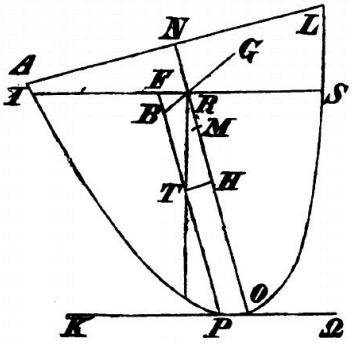
6. *ROK*] „*rωK*“ Tartalea. 7. perpendicularis] scrib. et perp. *TR*] om. Comm. „*ko*“ Tartalea. 8. *Ω*] „*o*“ Tartalea. 10. „ferret“ Tartalea. 12. „producta“ Tartalea. 21. aequae] >: aequalis; „aeque“ Tartalea.

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1881 Latin · printed p. 380 · public domain

ad axem, dimissa in humido ita, ut basis ipsius non tangat humidum, posita inclinata, non manet inclinata, sed restituetur in rectum.

esto portio rectanguli conoidalis, qualis dicta est, et dimissa in humidum, si est possibile, sit non recta, sed sit inclinata. secta autem ipsa per axem plano recto ad superficiem humidi portionis quidem sectio sit rectanguli coni sectio [*περὶ κων.* 11] quae $APOL$, axis autem portionis et diameter quae NO, superficiei autem humidi sectio sit IS. si igitur portio non est recta, non faciet quae NO ad IS angulos aequales. ducatur autem quae $K\Omega$ contingens sectionem rectanguli coni penes P, aequidistans autem ipsi IS. a P autem aequedistanter ipsi ON ducatur quae PF, et accipiantur centra grauitatum, et erit solidi quidem $APOL$ centrum R, eius autem, quod inter humidum, centrum B, et copuletur BR et educatur ad G, et sit solidi, quod supra humidi, centrum grauitatis G [*ἐπιπ. ἰσορρ.* I, 8]. et quoniam quae NO ipsius quidem RO est emiolia[1]), eius autem, quae usque ad axem, est maior



1) Nam centrum grauitatis conoidis rectanguli ita in axi

1. axem] addendum: ad tetragonum quod ab axe. 4. „rectangula“ Tartalea. 11. diameter] sc. sectionis. 19. quae] „que“ Tartalea. 20. „contra grauitum“ Tartalea. 21. *inter*] ↄ: intra. 22. BR] „*gtr*“ Tartalea. 23. quod supra *humidi*] *τὸ ὑπὲρ τοῦ ὑγροῦ*.

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Retained transcriptions and provenance

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Sources and rights

```
Zygon / It floats. Will it stay upright?

Ancient work: Archimedes, On Floating Bodies II.2 and II.4.

Greek transcriptions and processed manuscript images:
Archimedes Palimpsest Digital Release, hosted by University of Pennsylvania Libraries.
https://archimedespalimpsest.net/
Primary transcription: Reviel Netz and Nigel Wilson.
Sponsor: The Owner of the Archimedes Palimpsest.
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with Pillow LANCZOS, JPEG quality 82, subsampling 2. No crop, recolouring,
retouching or new decipherment. Published line coordinates are reused.
Greek excerpts select expanded abbreviations, join split words and retain
supplied, uncertain and sic flags. Accents/spacing/punctuation are editorial.
New English and contextual notes are teaching aids by Anton Soloviev (2026),
without independent philological review. No new Latin translation is supplied.

Latin witness:
Archimedis Opera omnia cum commentariis Eutocii, edited by J. L. Heiberg,
volume II, Leipzig: B. G. Teubner, 1881, printed pp. 376-377 and 379-380.
https://archive.org/details/archimedisopera04eutogoog
https://archive.org/download/archimedisopera04eutogoog/archimedisopera04eutogoog.pdf
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This is the pre-palimpsest edition: its Latin proof of II.2 breaks off,
and II.4 omits a denominator that Heiberg supplies in a footnote.
The retained Latin excerpts follow the body text, including that omission.

Modern mathematical comparison:
Kurt Girstmair and Gerhard Kirchner, Towards a Completion of Archimedes'
Treatise on Floating Bodies (2007), section 1, arXiv:0706.0119.
https://arxiv.org/abs/0706.0119
The analytic dry-base cap/centre conditions are independently derived in
docs/float-model.md. The paper is a reference, not redistributed content.
The workshop does not solve the paper's base-crossing extension.
```

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